

Design and Model Summary Report

Design Overview

DESIGN Simplex Lattice	RUNS 7	REPLICATES 0
CENTER POINTS 1	MIXTURE COMPONENTS 3	

Mixture Components

Component	Lower Bound	Upper Bound
X1	0	1
X2	0	1
X3	0	1

Design matrix

The complete design matrix in actual factor units is provided in Appendix A.

Model Structure

Responses: Y1, Y2

Included Terms

Intercept	Not included
Linear Terms	X1, X2, X3
2-Term Int.	X1 : X2, X1 : X3, X2 : X3
Quadratic Terms	Not included
3-Term Int.	X1 : X2 : X3

Model Matrix Diagnostics

Metric	Value
Observations	7
Columns	7
Rank	7
Aliased columns	0
Intercept	Yes
Regression DF	6
Residual DF	0
Status	Full rank, Saturated

Saturated model

The model has no residual degrees of freedom, so residual variance and related tests are not interpretable.

General Diagnostics

Variance Inflation Factors (VIF)

VIF < 5 is low, 5-10 is elevated, and VIF > 10 indicates high multicollinearity. For mixture terms these thresholds depend on parameterization; rank and alias diagnostics are primary.

Term	VIF	Flag
X1	1.6111	Low
X2	1.6111	Low
X3	1.6111	Low
X1:X2	1.3492	Low
X1:X3	1.3492	Low
X2:X3	1.3492	Low
X1:X2:X3	1.3968	Low

Leverage

Average leverage: 1

Run	Leverage
1	1
2	1
3	1
4	1
5	1
6	1
7	1

Response Conditions

Optimization limits and goals configured for each response.

Response	Lower Limit	Upper Limit	Goal
Y1	50	No	Maximize
Y2	50	No	Maximize

Response Analysis: Y1

Model Metrics

R^2	Adj. R^2	Q^2 (LOO)	PRESS (LOO)	RMSE	RMSE (LOO)
1	N/A	N/A	N/A	3.94e-14	N/A

Coefficient Summary

Term	Coefficient	Std. Error	95% CI half-width	Lower CI	Upper CI	p-value
X1	31	N/A	N/A	N/A	N/A	N/A
X2	113	N/A	N/A	N/A	N/A	N/A
X3	38	N/A	N/A	N/A	N/A	N/A
X1:X2	-120	N/A	N/A	N/A	N/A	N/A
X1:X3	18	N/A	N/A	N/A	N/A	N/A
X2:X3	-22	N/A	N/A	N/A	N/A	N/A
X1:X2:X3	354	N/A	N/A	N/A	N/A	N/A

Model F-test

F-value	F-crit (95%)	F-crit (99%)	p-value
N/A	N/A	N/A	N/A

Not interpretable

The F-test is undefined, commonly because model or residual degrees of freedom are insufficient.

Lack-of-Fit F-test

Not applicable

No replicated-run lack-of-fit test is applicable.

ANOVA

SD = sqrt(MS). Only the Residuals row reports residual standard deviation; RMSE = sqrt(SS_res / n).

Source	SS	dof	MS	SD
Total	4894.8571	6	815.8095	28.5624
Regression	4894.8571	6	815.8095	28.5624
Residuals	1.09e-26	0	N/A	N/A

Replicate Summary

Not applicable

No replicated runs are present.

Observed and Predicted Responses

Run	Observed	Predicted
1	38	38
2	70	70
3	113	113
4	39	39

Run	Observed	Predicted
5	42	42
6	31	31
7	60	60

Response Analysis: Y2

Model Metrics

R^2	Adj. R^2	Q^2 (LOO)	PRESS (LOO)	RMSE	RMSE (LOO)
1	N/A	N/A	N/A	2.85e-14	N/A

Coefficient Summary

Term	Coefficient	Std. Error	95% CI half-width	Lower CI	Upper CI	p-value
X1	44	N/A	N/A	N/A	N/A	N/A
X2	26	N/A	N/A	N/A	N/A	N/A
X3	52	N/A	N/A	N/A	N/A	N/A
X1:X2	24	N/A	N/A	N/A	N/A	N/A
X1:X3	76	N/A	N/A	N/A	N/A	N/A
X2:X3	-36	N/A	N/A	N/A	N/A	N/A
X1:X2:X3	195	N/A	N/A	N/A	N/A	N/A

Model F-test

F-value	F-crit (95%)	F-crit (99%)	p-value
N/A	N/A	N/A	N/A

Not interpretable

The F-test is undefined, commonly because model or residual degrees of freedom are insufficient.

Lack-of-Fit F-test

Not applicable

No replicated-run lack-of-fit test is applicable.

ANOVA

SD = sqrt(MS). Only the Residuals row reports residual standard deviation; RMSE = sqrt(SS_res / n).

Source	SS	dof	MS	SD
Total	1236	6	206	14.3527
Regression	1236	6	206	14.3527
Residuals	5.67e-27	0	N/A	N/A

Replicate Summary

Not applicable

No replicated runs are present.

Observed and Predicted Responses

Run	Observed	Predicted
1	52	52
2	30	30
3	26	26
4	67	67

Run	Observed	Predicted
5	41	41
6	44	44
7	55	55

Appendix A - Design Matrix

Run	X1	X2	X3
1	0	0	1
2	0	0.5	0.5
3	0	1	0
4	0.5	0	0.5
5	0.5	0.5	0
6	1	0	0
7	0.3333	0.3333	0.3333